Scalability & Future Plan

Date	15 March 2024
Team ID	UNM-293
Project Name	AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Gemini free tier quota 60 req/min	Limits concurrent users	High
2	Image gen latency 3-5 sec	Slows UX for many panels	High
3	No user authentication / DB	No multi-user history	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade
1	User Load	Handles 50 concurrent users	Use Redis queue + Load balancer + Paid Gemini tier
2	Data Storage	Session storage only	Firebase / MongoDB for user comics
3	Performance	Avg 1.4s story, 4s image	Caching, CDN, async batch image gen
4	Security	API key in .env	OAuth, secrets manager, user auth

Future Roadmap:

Phase	Planned Feature	Timeline	Expected Impact
Phase 1	Voice narration + Audio comics	Q2 2024	Accessibility + engagement
Phase 2	Multi-language comics (Tamil, Hindi)	Q3 2024	Wider audience
Phase 3	User accounts + Marketplace to share comics	Q4 2024	Community growth
Phase 4	Animation of comic panels + Video export	Q1 2025	Premium feature